0580_w24_ms_41_Q10

Question	Answer	Marks	Partial Marks	Guidance
10(a)(i)	$2x^2 + 5x - 187$ [= 0]	**M2**	**M1** for $(2x + 3)(x + 1) = 190$	Condone $2x^2 + 5x + 3 = 190$ for M1
	$(2x - 17)(x + 11)$ [= 0] oe	**M1**		FT *their* 3-term quadratic equation Allow $2x - 17 = 0$ and $x + 11 = 0$,or the correct answers after a partial factorisation seen, to imply M1 Accept use of quadratic formula, completing the square $\frac{-5\pm\sqrt{5^2-4\times2\times-187}}{2\times2}$ oe $\frac{-5\pm\sqrt{1521}}{4}$ at best **No recovery for short line or short root unless seen or implied in working** Fraction line must underscore the ± sign Length of root must cover – sign, ie reach the 4. Accept + for ± in formula $x = \frac{5}{4} \pm \sqrt{\frac{25}{16} + \frac{187}{2}}$ M1M1 implies M2M1
	Leading to $x = 8.5$ with no errors	**A1**		A1dep on M2M1 Must not include negative solution unless negative is clearly then rejected in the process, eg by crossing -11 out. A ring around 8.5 is not enough. No errors in any intermediate steps shown but after M1 awarded do not count eg short division line as an error. After M2 and M1 awarded condone answer 8.5 alone with no intermediate steps for A1

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Question	Answer	Marks	Partial Marks	Guidance
10(a)(ii)	59	**2**	**M1** for $6 \times 8.5 + 8$ oe or $6x + 8$ oe or **B1** for 9.5 and 20	May be on diagram Condone answer 29.5 for B1
10(b)(i)	$\frac{50}{360}\pi \times r^2 - \frac{1}{2}r^2 \times \sin 50 = 30$ oe	**M3**	**M1** for $\frac{50}{360}\pi \times r^2$ **M1** for $\frac{1}{2}r^2 \times \sin 50$ oe	implied by $0.0533..r^2$ [= 30] For method marks accept 3.14 or 22/7 for π Eg implied by $\frac{55}{126}r^2, \frac{157}{360}r^2$, (0.436 to 0.437)$r^2$ $\frac{50}{360}\pi \times 23.7^2$ is M0 Eg implied by 0.383[0…..]r^2 $\frac{1}{2} \times 23.7^2 \times sin50$ is M0
	23.70[9] to 23.72…	**A1**	must see at least 4 sig figs	Not 23.7 alone

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Question	Answer	Marks	Partial Marks	Guidance
10(b)(ii)	40.7 or 40.8 or 40.71 to 40.75…	**4**	**M2** for 2 × 23.7 × sin 25 oe or $\sqrt{23.7^2+23.7^2-2\times23.7\times23.7\cos50}$ oe or $\frac{23.7sin50}{\sin(\frac{180-50}{2})}$ oe or **M1** for $\frac{x}{23.7}=\sin25$ oe or for $23.7^2+23.7^2-2\times23.7\times23.7\cos50$ oe or $\frac{AB}{sin50}=\frac{23.7}{\sin(\frac{180-50}{2})}$ oe **AND** **M1** for $\frac{50}{360}\times2\times\pi\times23.7$ oe	Allow more accurate value for 23.7 Implied by 20.0 to 20.1, but 20 alone does not imply method. where x = ½ AB, 10 to 10.03 401.0 to 402.1 Implied by 20.67 to 20.71 isw eg +2 x 23.7